DATA ANALYTICS PROJECT

Excel · MySQL · Tableau · Power BI

Banking Analytics

Bank Loan Analytics & Credit-Debit Analytics

Final Project Report

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Chapter 1

Project Overview

1.1 About the Project

Banks hold large volumes of lending and transaction data, but the value is only realised when that data is turned into numbers a manager can read at a glance. This project builds that bridge for two datasets: a **Bank Loan** portfolio and a **Credit-Debit** transaction record. For each dataset the team defined a set of KPIs, wrote SQL queries in MySQL Workbench, and built dashboards in Excel, Tableau and Power BI.

Bank Loan Analytics looks at lending quality: how much was funded, how much has come back, how much interest was earned, and how much of the portfolio is delinquent, defaulted or unverified. Credit-Debit Analytics looks at cash flow: how credits compare with debits, how balances and transaction volume spread across banks and branches, and how often suspicious activity appears.

1.2 Headline Metrics

Bank Loan Analytics

65,535	₹750.97M	₹732.70M	₹155.29M
Total Loans	Total Loan Amount	Total Funded Amount	Total Interest

Credit-Debit Analytics

100,000	₹127.60M	₹127.29M	₹526.08M
Transactions	Total Credit	Total Debit	Total Balance

1.3 Project Roadmap

The project followed a five-step workflow from raw data to checked dashboards:

- **Step 1: Excel** – Data cleaning, KPI calculations and an initial pivot-based dashboard for each project.

- **Step 2: MySQL Workbench** – Both files loaded into the `banking_analytics_project2` schema; KPI and scenario queries written and run.
- **Step 3: Tableau** – Connected to MySQL; interactive dashboards with filters for each project.
- **Step 4: Power BI** – Connected to MySQL; interactive dashboards with slicers for each project.
- **Step 5: Quality Assurance** – Dashboard figures compared with the Excel KPI sheets and the database records.

Chapter 2

Dataset Overview

Two Excel files were supplied, one per project. Both were cleaned and then loaded into MySQL so that Tableau and Power BI could read the same source tables.

2.1 Source Files and Database Tables

Project	Source file	MySQL table	Rows	Columns
Bank Loan Analytics	Bank_Data_Analystics.xls	banking_data	65,535	52
Credit-Debit Analytics	Debit_and_Credit_bankin	debit_credit_transacti	100,000	12

Row counts were confirmed in MySQL with `SELECT COUNT(*)` on each table (65,535 and 100,000), and they match the totals shown on the dashboards.

2.2 Bank Loan Data – Key Fields

Group	Fields
Borrower	Client Id, Client Name, Age, Date of Birth, Gender ID, Caste, Religion, Home Ownership
Loan terms	Loan Amount, Funded Amount, Funded Amount Inv, Term, Int Rate, Grade, Sub Grade, Product Code, Purpose Category
Repayment	Total Payment, Total Rec Principal, Total Rec Interest, Total Rec Late Fee, Recoveries, Collection Recovery Fee
Status and risk	Loan Status, Is Delinquent Loan, Is Default Loan, Delinq 2 Yrs, Verification Status
Location and staff	State Name, City, Region Name, Branch Name, Bank Name, Credit Officer Name, BH Name
Dates	Disbursement Date, Disbursement Date (Years), Closed Date, Loan Transfer Date, Next Meeting Date

2.3 Credit-Debit Data – Key Fields

Field	Description
Customer ID, Customer Name	Identifies the customer making the transaction
Account Number	Account that holds the balance
Transaction Date	Date of the transaction (calendar year 2024)
Transaction Type	Credit or Debit
Amount, Balance, Currency	Transaction value, account balance after it, and currency
Description	Free-text description of the transaction
Transaction Method	Credit Card, Debit Card or Bank Transfer
Branch, Bank Name	Six branches and six banks: Axis, HDFC, ICICI, Kotak Mahindra, Punjab National and State Bank of India

Chapter 3

Objectives and KPIs

The objective of each project is to answer a short list of business questions with numbers. The KPIs below were calculated in Excel and then reproduced in SQL, Tableau and Power BI. Amounts follow the Excel KPI sheets, with a millions (M) equivalent where it makes the figure easier to read.

3.1 Bank Loan KPIs

KPI	Value	Definition
Total Loans	65,535	Number of loan records in the cleaned dataset
Total Loan Amount	₹ 75,09,67,500.00 (about ₹750.97M)	Total amount of loans approved/issued across all loan records
Total Funded Amount	₹ 73,26,97,200.00 (about ₹732.70M)	Total amount actually funded to borrowers
Total Interest	₹ 15,52,88,805.82 (about ₹155.29M)	Total interest amount recorded across all loans
Total Recoveries	₹ 65,22,808.07 (about ₹6.52M)	Total amount recovered from loans, including repayments/recovery amounts
Total Collection Recovery Fee	₹ 7,56,410.80 (about ₹0.76M)	Total fees associated with collection/recovery activities
Default Loans	1,020	Number of loans identified as default loans
Delinquent Loans	7,106	Number of loans identified as delinquent/overdue
Not Verified Loans	16,921	Number of loans where the verification status is Not Verified
Default Rate	1.6%	Percentage of total loans that are in default
Delinquent Rate	10.8%	Percentage of total loans that are delinquent
Not Verified Rate	25.8%	Percentage of total loans that are not verified
Average Loan Amount	₹ 11,459.03 (about ₹11.46K)	Average loan amount per loan record
Average Interest Rate	12.0%	Average interest rate charged across the loan portfolio
Loan Status Count	9	Number of loans by current loan status
Loan Maturity	43.2 months	Analysis of loan maturity/repayment period
Loan Verification Rate	74%	Percentage of loans that are verified

KPI	Value	Definition
Total Collection Recovery	₹ 65,22,808.07 (about ₹6.52M)	Overall recovery performance from the loan portfolio

Besides these calculated KPIs, the loan dashboards show branch-wise performance, state-wise loans, religion-wise loans, product-group loans, disbursement trend, grade-wise loans, loan status split and age-group loans.

3.2 Credit-Debit KPIs

KPI	Value	Definition
Total Customers	1,00,000	Total number of unique customers/accounts in the dataset
Total Transactions	1,00,000	Total number of transaction records in the dataset
Total Credit Amount	₹ 12,76,03,386.41 (about ₹127.60M)	Total amount of transactions where Transaction Type = Credit
Total Debit Amount	₹ 12,72,85,269.22 (about ₹127.29M)	Total amount of transactions where Transaction Type = Debit
Total Balance	₹ 52,60,78,736.26 (about ₹526.08M)	Total account balance recorded across all customer accounts
Credit to Debit Ratio	1.00x	Total Credit Amount ÷ Total Debit Amount
Net Transaction Amount	₹ 3,18,117.19 (about ₹0.32M)	Total Credit Amount − Total Debit Amount; measures net cash flow
Account Activity Ratio	0.0002 (0.02%)	Number of transactions ÷ account balance; customer activity relative to balance
Transactions per Day/Week/Month	297.62	Number of transactions per period; shows transaction-volume trends
Total Transaction Amount by Branch	₹ 4,29,11,469.79 (about ₹42.91M)	Sum of transaction amounts grouped by branch (value as shown in the KPI sheet)
Transaction Volume by Bank	6	Sum of transaction amounts grouped by bank name; compares banks
Transaction Method Distribution	3	Transactions grouped by method; shows usage of each method
Branch Transaction Growth	-96.57%	Percentage change in transaction amount at each branch over a defined period
High-Risk Transaction Flag	5,000	Flags transactions above a predefined amount or with irregular activity
Suspicious Transaction Frequency	5.00%	Count of flagged high-risk transactions over a specified period

Chapter 4

Process and Methodology

The same steps were followed for both projects so that the results could be compared across tools.

4.1 Workflow

Step	Tool	What was done
1	Excel	Cleaned the raw sheets, calculated the KPIs, and built a pivot-based dashboard with slicers.
2	MySQL Workbench	Created the schema `banking_analytics_project2`, loaded `banking_data` and `debit_credit_transactions`, and wrote KPI and scenario queries.
3	Tableau	Connected to MySQL, imported the tables, built calculated fields and one interactive dashboard per project.
4	Power BI	Connected to MySQL, imported the tables, built measures and one interactive report per project.
5	Quality assurance	Compared dashboard values with the Excel KPI sheets and database counts; noted differences (see Chapter 9).

4.2 Connecting the BI Tools to MySQL

Tableau and Power BI were both connected to MySQL for importing the data, so every tool reads from the same two tables. This keeps the figures consistent and means a change in the database flows through to the dashboards after a refresh.

Chapter 5

SQL Analysis in MySQL Workbench

The SQL phase confirmed the data load and turned business questions into queries. Part 1 of the loan script holds the KPI queries from the project presentation; Part 2 adds extra real-world analyst scenarios. The credit-debit script follows the same pattern.

5.1 Bank Loan Queries

The first query counts the rows in `banking_data` and returns 65,535, matching the dashboard. Scenario 2 is a loan-officer leaderboard that answers: *which loan officers manage the most volume, and do any of them have a default rate high enough to warrant a review?*

```
SELECT credit_officer_name,
       COUNT(*)           AS loans_managed,
       SUM(loan_amount)   AS total_loan_amount,
       SUM(CASE WHEN is_default_loan = 'Y' THEN 1 ELSE 0 END) AS default_count,
       ROUND(100.0 * SUM(CASE WHEN is_default_loan = 'Y' THEN 1 ELSE 0 END)
             / COUNT(*), 2) AS default_rate_pct
FROM banking_data
GROUP BY credit_officer_name
HAVING COUNT(*) >= 20          -- ignore officers with too few loans
ORDER BY default_rate_pct DESC
LIMIT 20;
```

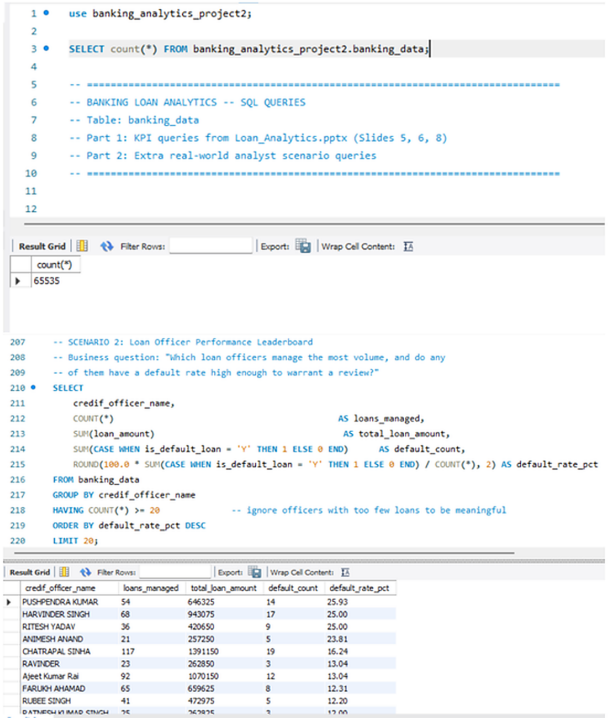


Figure 5.1: MySQL Workbench – row count and loan officer leaderboard (Scenario 2).

Result (top of the list)

Credit officer	Loans managed	Total loan amount	Defaults	Default rate
Pushpendra Kumar	54	646,325	14	25.93%
Harvinder Singh	68	943,075	17	25.00%
Ritesh Yadav	36	420,650	9	25.00%
Animesh Anand	21	257,250	5	23.81%
Chatrapal Sinha	117	1,391,150	19	16.24%
Ravinder	23	262,850	3	13.04%
Ajeet Kumar Rai	92	1,070,150	12	13.04%
Farukh Ahamad	65	659,625	8	12.31%
Rubee Singh	41	472,975	5	12.20%

Four officers have a default rate of 23% or more, against 1.56% for the whole portfolio. Pushpendra Kumar tops the list at 25.93% across 54 loans, so these officers are the first candidates for a portfolio review.

5.2 Credit-Debit Queries

The first query counts the rows in `debit_credit_transactions` and returns 100,000. Scenario 7 measures **bank market share by transaction volume**: what percentage of total transaction value each bank represents.

```sql
SELECT bank_name,
       SUM(amount) AS total_amount,
       ROUND(100.0 * SUM(amount) /
             (SELECT SUM(amount) FROM debit_credit_transactions), 2) AS market_share_pct
FROM debit_credit_transactions
GROUP BY bank_name
ORDER BY market_share_pct DESC;
```

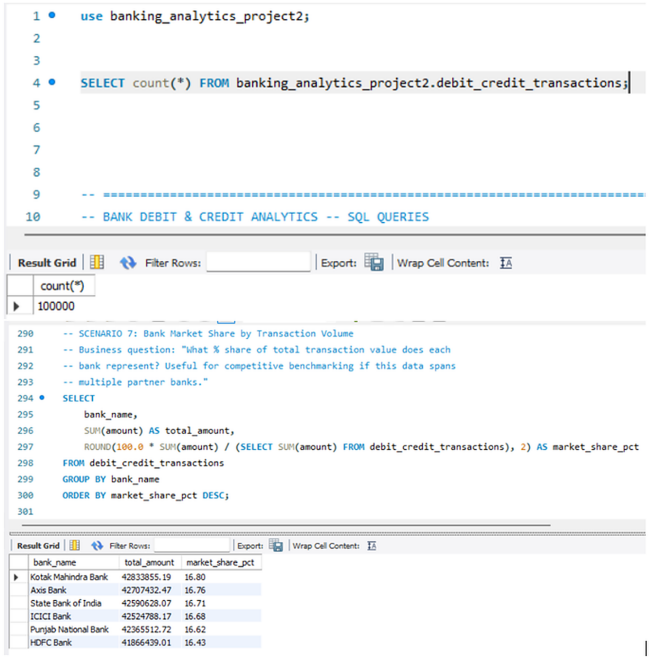


Figure 5.2: MySQL Workbench – row count and bank market share (Scenario 7).

Bank	Total amount (₹)	Market share
Kotak Mahindra Bank	42,833,855.19	16.80%
Axis Bank	42,707,432.47	16.76%
State Bank of India	42,590,628.07	16.71%
ICICI Bank	42,524,788.17	16.68%
Punjab National Bank	42,365,512.72	16.62%
HDFC Bank	41,866,439.01	16.43%

The six banks sit within 0.4 percentage points of each other, and the six totals add up to about ₹254.9M, the same figure as Total Transactions on the Power BI dashboard.

Chapter 6

Dashboards and Visual Analysis

Six dashboards were built, one per project in each of Excel, Tableau and Power BI. Each tool shows the same data with its own strengths: Excel for speed, Tableau and Power BI for interactivity.

6.1 Excel Dashboards

The Excel dashboards are pivot-based and use slicers to filter the whole page at once.

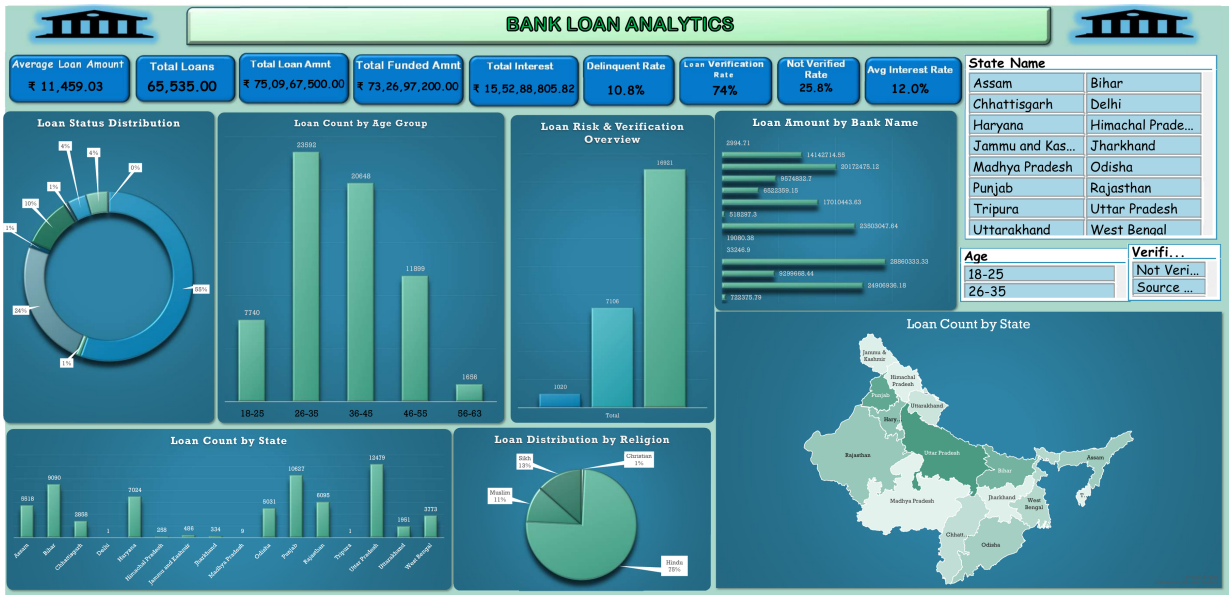


Figure 6.1: Excel – Bank Loan Analytics dashboard.

Visualizations used

- **KPI cards** – Average loan amount, total loans, loan amount, funded amount, interest, delinquent rate, verification rate, not-verified rate, average interest rate
- **Doughnut chart** – Loan status distribution
- **Column charts** – Loan count by age group and by state; loan risk and verification overview
- **Bar chart** – Loan amount by bank name
- **Map** – Loan count by state
- **Pie chart** – Loan distribution by religion
- **Slicers** – State name, age, verification status

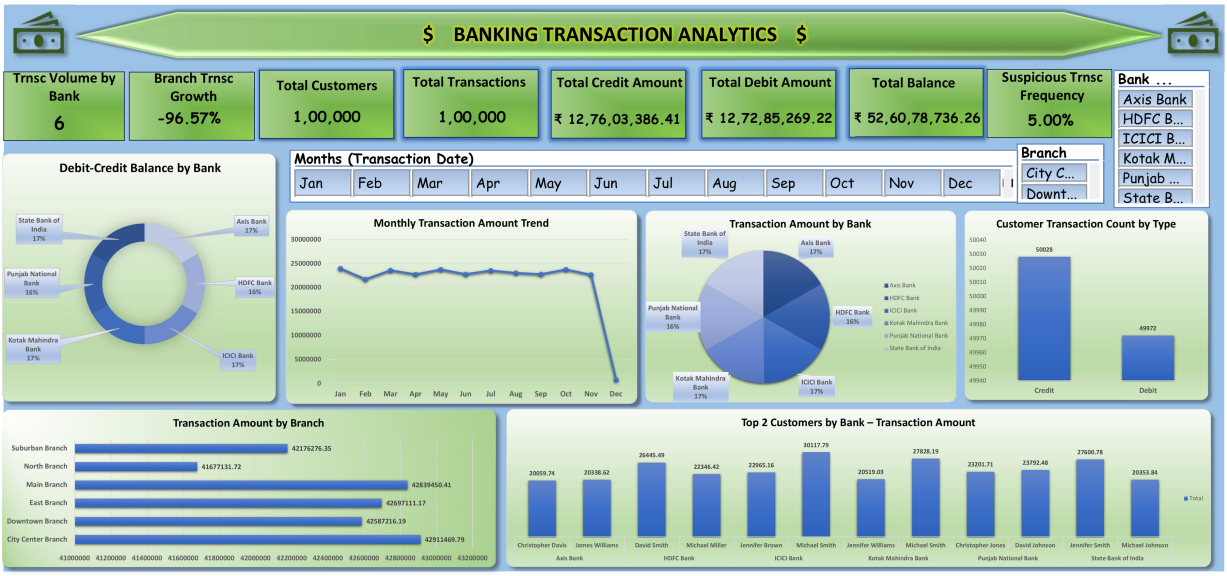


Figure 6.2: Excel – Banking Transaction Analytics dashboard.

Visualizations used

- **KPI cards** – Transaction volume by bank, branch transaction growth, total customers, total transactions, credit, debit, balance, suspicious transaction frequency
- **Doughnut and pie charts** – Debit-credit balance by bank; transaction amount by bank
- **Line chart** – Monthly transaction amount trend
- **Column chart** – Customer transaction count by type (credit 50,028; debit 49,972)
- **Bar chart** – Transaction amount by branch
- **Column chart** – Top 2 customers by bank
- **Slicers** – Bank, branch and transaction month

6.2 Tableau Dashboards

Both Tableau dashboards read from MySQL and use filters on the left or right edge of the page.

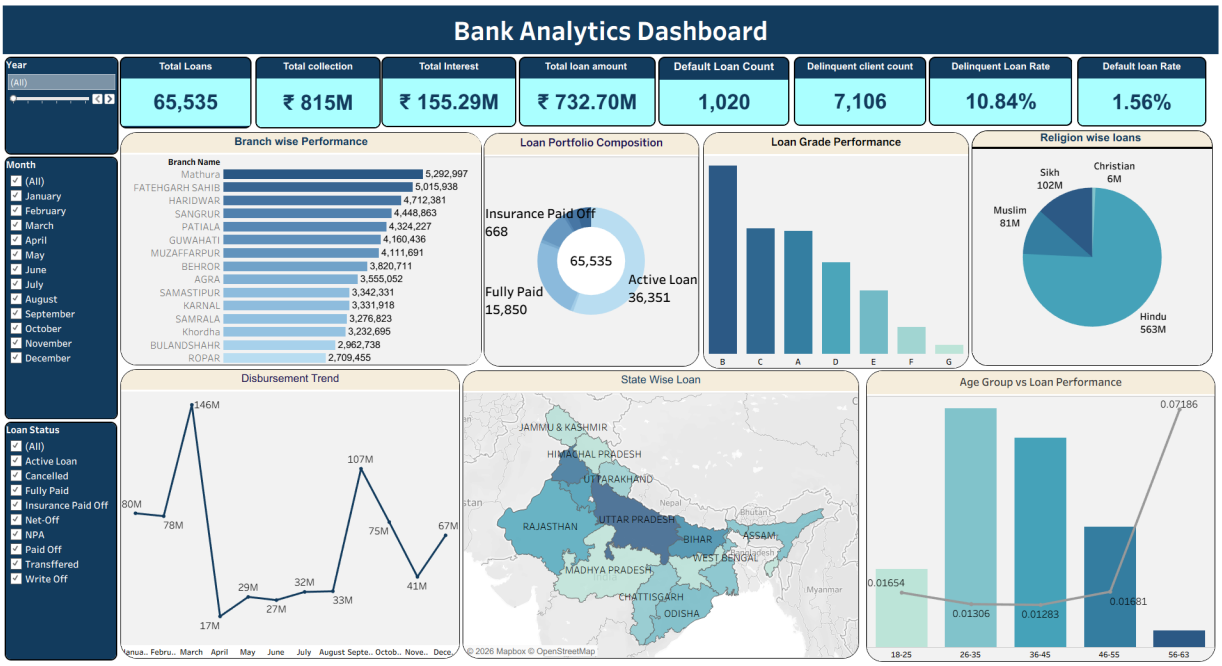


Figure 6.3: Tableau – Bank Loan Analytics dashboard.

Visualizations used

- **KPI cards** – Total loans 65,535; total collection ₹815M; total interest ₹155.29M; total loan amount ₹732.70M; default loan count 1,020; delinquent client count 7,106; delinquent loan rate 10.84%; default loan rate 1.56%
- **Bar chart** – Branch-wise performance
- **Doughnut chart** – Loan portfolio composition (active, fully paid, insurance paid off)
- **Column chart** – Loan grade performance
- **Pie chart** – Religion-wise loans
- **Line chart** – Disbursement trend by month
- **Filled map** – State-wise loan
- **Column + line** – Age group versus loan performance
- **Filters** – Year, month, loan status

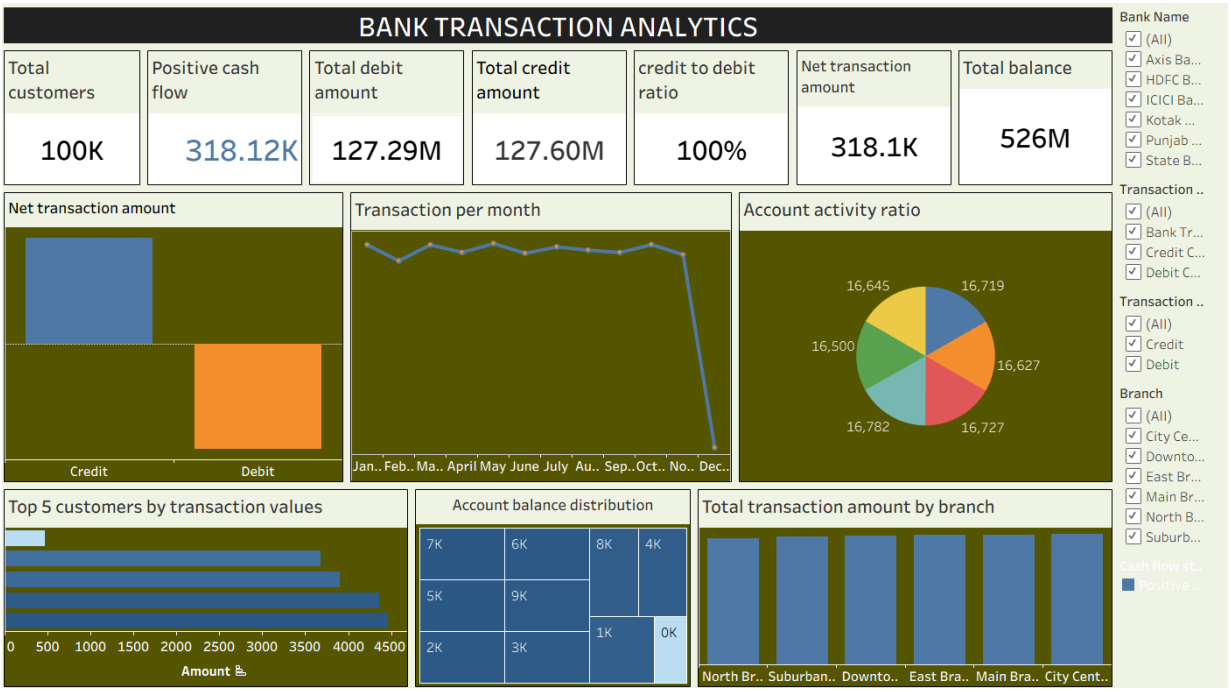


Figure 6.4: Tableau – Bank Transaction Analytics dashboard.

Visualizations used

- **KPI cards** – Total customers 100K; positive cash flow 318.12K; total debit 127.29M; total credit 127.60M; credit-to-debit ratio 100%; net transaction amount 318.1K; total balance 526M
- **Bar chart** – Net transaction amount (credit versus debit)
- **Line chart** – Transactions per month
- **Pie chart** – Account activity ratio
- **Bar chart** – Top 5 customers by transaction value
- **Treemap** – Account balance distribution
- **Column chart** – Total transaction amount by branch
- **Filters** – Bank name, transaction method, transaction type, branch

6.3 Power BI Dashboards

The Power BI dashboards use dark themes with a KPI strip along the top and slicers for the main filters.

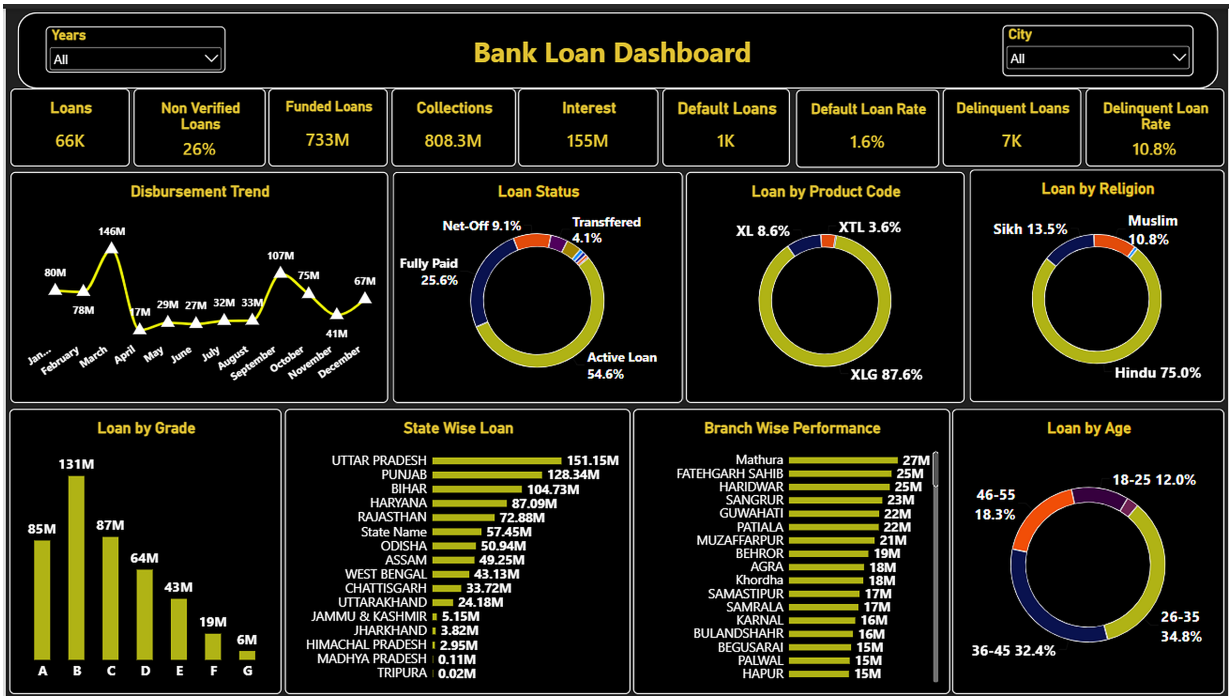


Figure 6.5: Power BI – Bank Loan Dashboard.

Visualizations used

- **KPI strip** – Loans 66K; non-verified loans 26%; funded loans 733M; collections 808.3M; interest 155M; default loans 1K; default loan rate 1.6%; delinquent loans 7K; delinquent loan rate 10.8%
- **Line chart** – Disbursement trend
- **Doughnut charts** – Loan status, loan by product code, loan by religion, loan by age
- **Column chart** – Loan by grade
- **Bar charts** – State-wise loan and branch-wise performance
- **Slicers** – Years, city

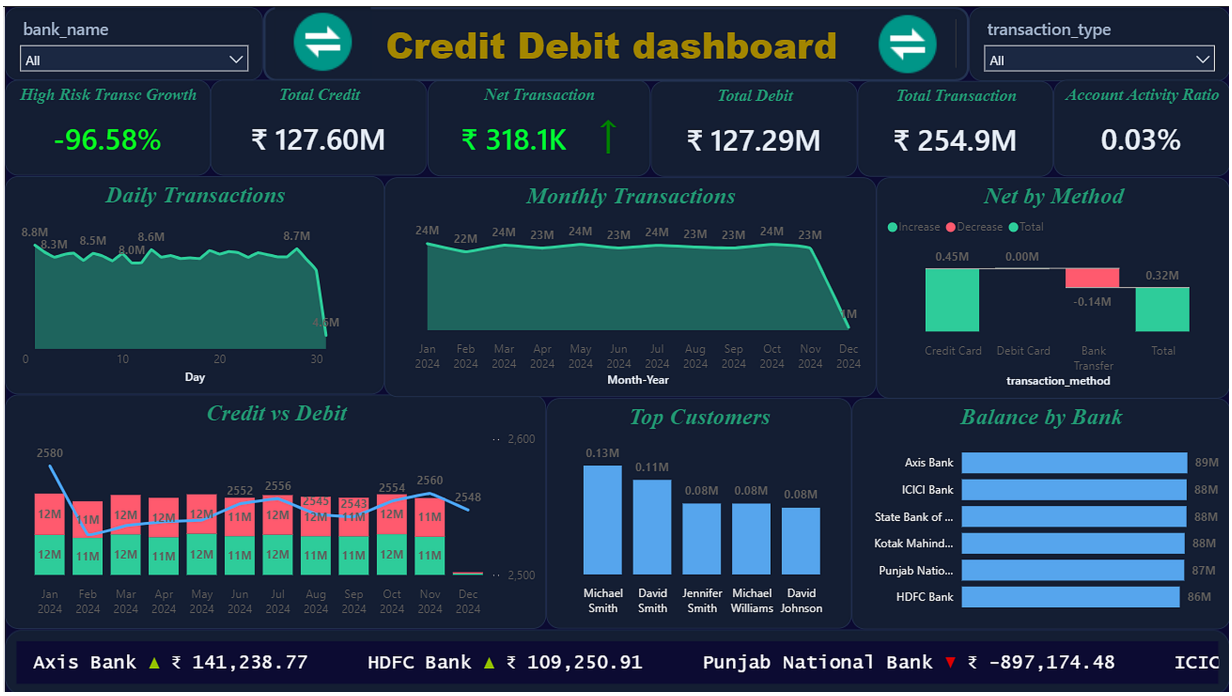


Figure 6.6: Power BI – Credit Debit Dashboard.

Visualizations used

- **KPI cards** – High-risk transaction growth -96.58%; total credit ₹127.60M; net transaction ₹318.1K; total debit ₹127.29M; total transaction ₹254.9M; account activity ratio 0.03%
- **Area charts** – Daily and monthly transactions
- **Waterfall chart** – Net by transaction method
- **Stacked column + line** – Credit versus debit by month
- **Column chart** – Top customers
- **Bar chart** – Balance by bank
- **Ticker** – Net position per bank
- **Slicers** – Bank name, transaction type

Chapter 7

Key Insights

7.1 Bank Loan Findings

1. **Funding is close to approval** – ₹732.70M of the ₹750.97M approved was actually funded, about 97.6%.
2. **Delinquency is the main risk** – 10.84% of loans (7,106) are delinquent, against 1.56% (1,020) in default. Overdue loans are about seven times more common than defaults.
3. **A quarter of loans are unverified** – 16,921 loans (25.8%) have a verification status of Not Verified, a compliance and credit-risk gap.
4. **Lending is concentrated in a few states** – Power BI shows Uttar Pradesh (₹151.15M), Punjab (₹128.34M), Bihar (₹104.73M), Haryana (₹87.09M) and Rajasthan (₹72.88M) at the top of state-wise lending.
5. **Grade B leads the portfolio** – Grade B holds ₹131M, followed by C (₹87M), A (₹85M) and D (₹64M); grades E, F and G together are smaller.
6. **Core borrowers are 26 to 45 years old** – The 26-35 group holds 34.8% of loans and 36-45 holds 32.4%; 18-25 holds 12.0%.
7. **Most loans are still active** – Active loans are 54.6% of the portfolio, fully paid 25.6%, net-off 9.1% and transferred 4.1%.
8. **Disbursement peaks in March** – Monthly disbursement peaks at about ₹146M in March and falls to about ₹17M in April, then recovers to ₹107M in September.
9. **A few loan officers carry high default rates** – Four officers with 20 or more loans have default rates of 23.8% to 25.9% (Chapter 5).

7.2 Credit-Debit Findings

1. **Credits and debits are almost equal** – ₹127.60M in credits against ₹127.29M in debits gives a credit-to-debit ratio of 1.00x. Net inflow is only ₹318.1K, about 0.25% of credits.
2. **Transactions split evenly by type** – Credit has 50,028 transactions and debit 49,972, so the book is close to balanced.
3. **No bank dominates** – Market share runs from 16.43% (HDFC) to 16.80% (Kotak Mahindra), and balances per bank are all between about ₹86M and ₹89M.
4. **Branch volumes are close** – City Center Branch is highest at ₹42.91M and North Branch lowest at ₹41.68M in the Excel dashboard.

5. **Net position differs by bank** – The Power BI ticker shows Axis Bank +₹141,238.77 and HDFC Bank +₹109,250.91, while Punjab National Bank is negative at -₹897,174.48.
6. **Suspicious activity is 5%** – 5,000 of the 100,000 transactions are flagged as high-risk.
7. **Monthly volume is steady until December** – Volume stays around ₹22M to ₹24M a month from January to November, then drops sharply in December. This looks like partial data for the month, which also explains the -96.57% branch transaction growth.

Chapter 8

Recommendations

The recommendations follow from the findings in Chapter 7. They are suggestions for discussion, not instructions.

8.1 Bank Loan

- **Strengthen early follow-up on overdue loans.** With delinquency at 10.84%, an early-warning process on loans that miss a first payment would cost less than recovering them later.
- **Verify before disbursing.** Move the 25.8% not-verified share down by making verification a condition of funding.
- **Watch state concentration.** Uttar Pradesh, Punjab and Bihar carry the largest lending; track their delinquency separately so a local problem does not become a portfolio problem.
- **Review high-default loan officers.** Start with the four officers above 23% and check whether the cause is the borrower mix, the area or the lending process.
- **Target the core segment.** Borrowers aged 26 to 45 and grades A to C make up most of the volume; marketing and underwriting can be tuned for them.

8.2 Credit-Debit

- **Monitor liquidity.** Net inflow is only 0.25% of credits, so a small rise in debits could turn the net position negative.
- **Review Punjab National Bank outflows.** It is the one bank with a clearly negative net position on the Power BI ticker.
- **Tighten suspicious-transaction alerts.** A 5% flag rate is high; refine the rule so that analysts review fewer, better-targeted cases.
- **Fix the December data gap.** Reload or confirm December before quoting growth rates, so the -96.57% figure does not mislead.
- **Use branch and bank comparisons carefully.** Differences are small, so look for trends over time rather than ranking on a single month.

Chapter 9

Challenges and Learnings

9.1 Challenges

- **Large files.** 65,535 loan rows with 52 columns and 100,000 transactions slowed Excel and needed careful cleaning.
- **Connecting to MySQL.** Setting up the Tableau and Power BI connections and importing the tables took trial and error.
- **Different totals between tools.** Total collection shows as ₹815M in Tableau, 808.3M in Power BI and 814,902,500 in an earlier Tableau view, so the calculation and filters differ slightly between tools.
- **Different labels for similar numbers.** The Tableau card named "Total loan amount" (₹732.70M) matches the Excel *Funded* amount, while the Excel Total Loan Amount is ₹750.97M.
- **Rounded cards.** Power BI shows Loans as 66K and Default Loans as 1K, rounded from 65,535 and 1,020.
- **Partial December data.** It distorts growth rates and the monthly trend lines in every tool.
- **A stray "State Name" row** appears in the Power BI state-wise chart and should be filtered out.

9.2 Learnings

- Clean data and a clear definition for every KPI come first; most differences between tools came from definitions, not from the data.
- A SQL layer keeps every tool on one source of truth and makes checking easy.
- Checking dashboards against the database early catches mismatches before they reach a presentation.
- Each tool has a role: Excel for fast first views, Tableau and Power BI for interactive dashboards.
- Teamwork across a 7-member group needs a shared naming convention for tables, measures and files.

Chapter 10

Tools and Technology Stack

The project used a standard analytics toolkit for data preparation, database analysis and visualization.

Tool	Role in the project
Excel	Data cleaning, KPI calculation, pivot tables, charts and the first dashboards
MySQL Workbench	Schema design, data loading, KPI queries and scenario queries
Tableau	Connection to MySQL, calculated fields and interactive dashboards
Power BI	Connection to MySQL, measures and interactive reports

Chapter 11

Conclusion

This project built a complete workflow for two banking datasets, from raw Excel files to a shared MySQL layer and six dashboards in Excel, Tableau and Power BI. The same KPIs were reproduced in each tool and checked against the database, which gave the team confidence in the main figures: 65,535 loans with ₹732.70M funded, ₹155.29M of interest, a 10.84% delinquency rate and a 1.56% default rate; and 100,000 transactions with ₹127.60M in credits against ₹127.29M in debits.

The analysis points to two clear messages. In lending, the risk sits in delinquency, unverified loans and a handful of high-default loan officers rather than in outright defaults. In transactions, money flows evenly across banks, branches and transaction types, with a very thin net inflow and a 5% share of flagged transactions worth a closer look. With the December data gap resolved and the KPI definitions aligned across tools, the dashboards can serve as a reliable monitoring view for both portfolios.

Team – Group 4

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Excel · SQL · Tableau · Power BI